Rent or Toxicity?

Strategic exploitability of learning market makers with public wallet information

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6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

Good morning. We are Deepu and Miguel, Example 4: an algorithmic market with adaptive agents. The title is a question, rent or toxicity, and by the end you will know why those two things are impossible to tell apart from the outside, and what that means for the people trading against professionals on a public exchange. Ten minutes: two on why, three on the model, four on what we found, one on what is next.

2026: the year retail leverage went wrong at national scale

−44%
KOSPI, June 19 to July 29

1.2 million
retail accounts margin-called: 3.4% of all adults

$39 billion
retail losses on single-stock leveraged ETFs

“Retail” means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- trading on a phone app, with borrowed money, against professionals they cannot see

President Lee Jae-myung, on X, October 14, 2026

살아야지요. 어떻게든 사셔야 합니다. 힘드시면 109번 전화를 하십시오.

사실 길, 살아야 하는 이유를 함께 찾아드릴 것입니다.

“You must live. Somehow, you must live. If it is hard, call 109. We will find, together, a way and a reason to live.”

Sources: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, “KOSPI Crash 2026”); Maeil Shinmun, Oct 22, 2026; x.com/Jaemyung_Lee/status/2110145832511225984. 109 is Korea’s suicide-prevention line.

Why we care. This summer the Korean stock market fell forty-four percent in six weeks, and July 28 was the worst single day on record. What made it a national event rather than a market event is who was holding the risk. The word for that is retail, and retail means everyday people: not banks, not funds. Ninety-two percent of the holders of these products were individuals. By mid-July 1.2 million retail accounts had been margin-called, which is three point four percent of every adult in the country, and three hundred thousand were force-liquidated. Citi puts retail losses on single-stock leveraged ETFs alone at thirty-nine billion dollars, and sixty-two percent of those accounts belonged to people in their twenties and thirties. Students, office workers, people with ordinary jobs, trading on a phone app with borrowed money against professionals they could not see. On October 14 the President of Korea posted this on X: you must live; if it is hard, call 109, the suicide-prevention line. That is the level at which this stopped being a finance story. Our project is about the gap between what one of those people can see and what the professional opposite them can see, and we think closing that gap is a financial-literacy problem as much as a market-design one.

What “leverage” is, and why a 5% move can take everything

You put in $1,000. At 20× leverage you control a $20,000 position.



yours: $1,000 **borrowed from the exchange**: $19,000

price +5%

position worth $21,000

you doubled your money: +$1,000

price −5%

position worth $19,000: exactly the loan

you lost everything: −$1,000

the exchange closes your position automatically (**liquidation**); in a crash, millions of these happen at once and push the price further down

A 2× ETF (Korea) does this for you at 2×. On a crypto perpetuals exchange the multiplier is a **slider**, up to 50×.

The yield is visible on the way in. The risk, and who is on the other side, is not.

Since nobody here needs to know this for a living: leverage is trading with borrowed money. You put in a thousand dollars; at twenty times leverage the exchange lends you nineteen thousand more and you control a twenty thousand dollar position. If the price goes up five percent, the position is worth twenty-one thousand; you pay back the nineteen, you keep two, you doubled your money. That is the yield people see. If the price goes down five percent, the position is worth nineteen thousand, which is exactly the loan. Your thousand is gone. The exchange does not wait for you: it closes the position automatically, which is called liquidation, and in a crash millions of these happen at once and push the price down further, which liquidates the next person. In Korea the product was a two-times ETF, a fund that does this for you at two times. On a crypto perpetuals exchange the multiplier is a slider that goes to fifty. The yield is visible on the way in; the risk, and who is on the other side taking your money, is not. That second part is what we study.

These products are moving toward US retail; Korea should be the lesson

In the US, right now

- **CLARITY Act** (crypto market structure): House 294 to 134, July 2025; one vote short in the Senate, Sept 15, 2026; revivable
- **CFTC**, Aug 2026: rules to let exchanges offer leveraged crypto trading to Americans; filed Sept 17
- **Hyperliquid** blocks US users today; its CEO is discussing a CFTC path

In Korea, after the crash

- new listings halted; minimum deposit tripled; exposure capped at 20% of assets. All of it *after* the losses.

On a public-wallet venue, every trade you make is on the tape

fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
public, forever; the professionals read it

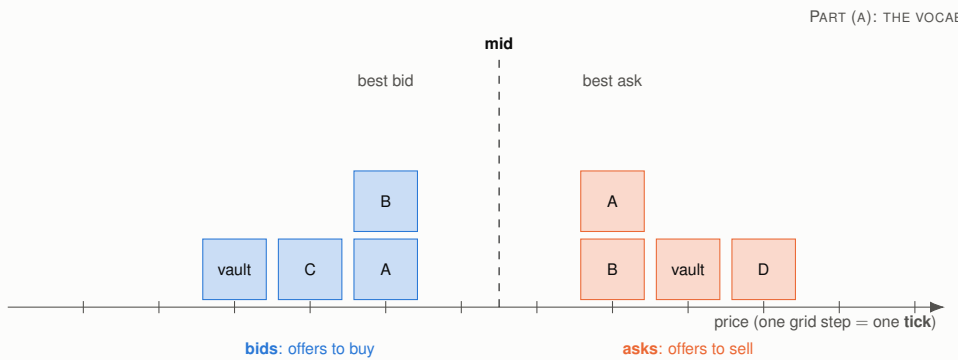
What an everyday trader would need to know

- what a trade really costs
- who is on the other side
- whether their identity is used against them
- which venue rules protect them in stress

Korea's protections came after the losses. We measure these four things now, on the real venue and in a model where the rules can be changed, so the protections can come first.

The products that did this in Korea are moving toward US retail, and the legislative machinery is in motion. The CLARITY Act, the crypto market-structure bill, passed the House two hundred ninety-four to one hundred thirty-four in July 2025; it failed a Senate procedural vote forty-nine to fifty on September 15 this year, but the no vote that sank it was cast specifically to preserve a path to bring it back. Separately, in August the CFTC chairman announced rules that would let exchanges offer leveraged crypto trading to Americans under CFTC oversight, and that rulemaking was filed in September. Hyperliquid, the venue we study, blocks US users today, but its CEO has been publicly discussing exactly that path. And on this kind of venue there is one more thing: every trade you make carries your wallet address on a public tape, forever, and the professionals on the other side read it. We will show that they do. Korea's response was real: new listings halted, the minimum deposit tripled, exposure capped at twenty percent of assets. But every one of those came after the losses. We think what happened in Korea should be read as a lesson, and the protections should come first. For that, someone has to measure what an everyday trader would need to know: what a trade really costs beyond the quoted price; who is on the other side and earns from them; whether their public identity is used against them; and which venue rules protect the uninformed in a stress episode, and which quietly make them pay. That is what this project measures, on the real venue and in a model where the rules can be changed.

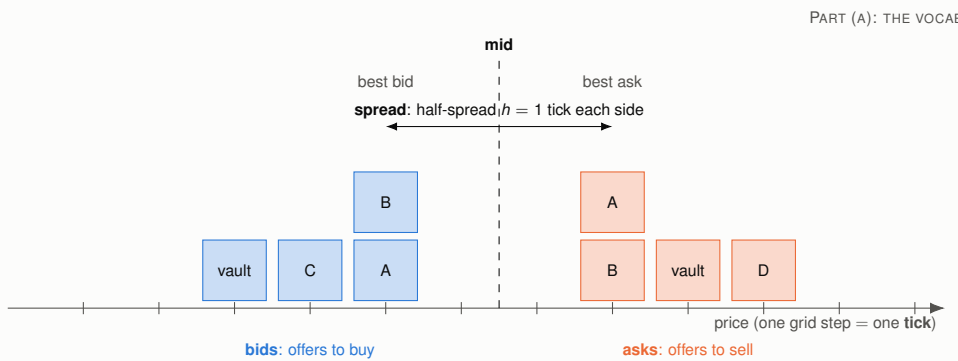
An exchange is a list of standing offers: the order book



Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

Vocabulary, because the audience has not seen this before. An exchange is a list of standing offers, the order book. Price lives on a grid; one grid step is a tick. On the left, in blue, are bids: offers to buy at that price. On the right, in orange, are asks: offers to sell. The people who post these standing offers are market makers; each box is one maker's order and on Hyperliquid each box carries the maker's public wallet address. The dashed line is the mid, halfway between the best bid and the best ask. [click] The distance between best bid and best ask is the spread; we write the half-spread h in ticks, here one tick each side. [click] A taker is someone who accepts an offer instead of posting one. T buys one unit: he hits the best ask, and that is a fill for maker B, because B was first in line at that price and A was behind. A bigger order takes A's too and walks to the next price, and that walking is most of what a trade really costs. What becomes public: the fill itself, buy, price, both wallets, immediately. What does not: whether B made or lost money on it. That only shows up as the price moves afterwards, which is the next slide.

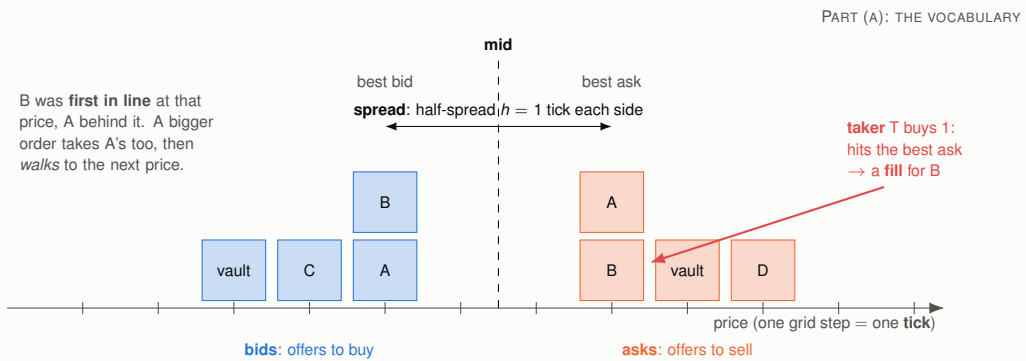
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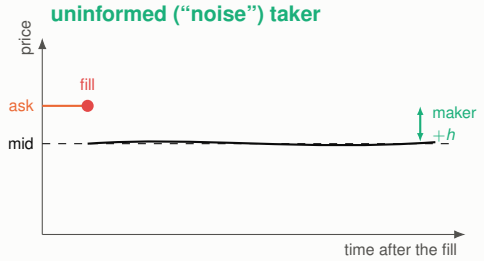


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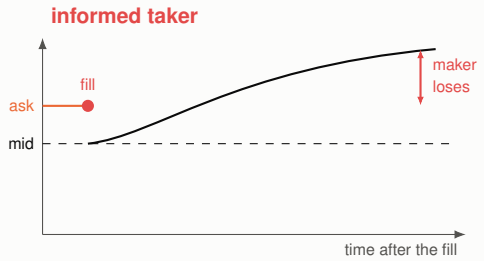
Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

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Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread



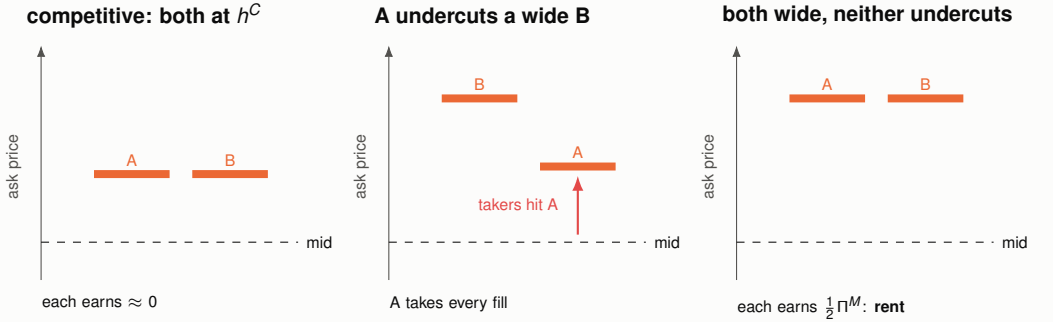
the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

How do we know whether the maker won on a fill? We look at where the price goes afterwards. On the left, the taker was uninformed, a noise trader; the price sits where it was, and the maker sold above the mid, so it keeps the half-spread. On the right the taker knew something; the price runs up past the price the maker sold at, and the maker has sold too cheap. That signed difference, mid a few seconds later minus the fill price from the maker's point of view, is the markout. It is the single measurement everything in our data work rests on. A stream of fills with negative markouts is what practitioners call toxic flow, and economists call adverse selection. A maker's whole business is being in line for the left-hand fills and not the right-hand ones.

Rent: what makers earn by not undercutting each other

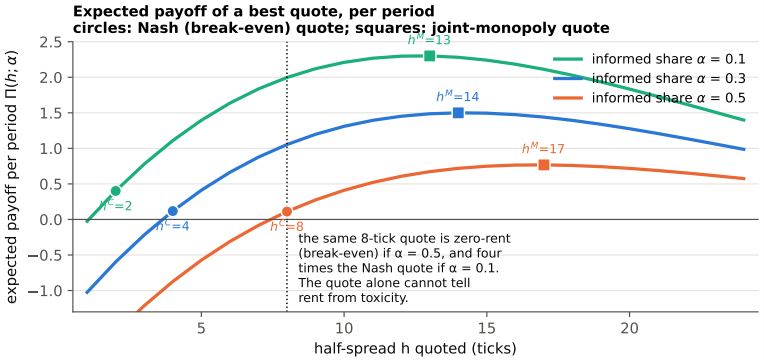


h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

The other word in the title: rent. Three pictures of the same two makers' ask prices. Left: both quote the competitive half-spread h-C, where what you earn from noise takers just covers what you lose to informed ones; each earns about zero. Middle: B quotes wide and A undercuts by a tick; takers always hit the tighter price, so A takes every fill and B gets nothing. That is why a wide quote is fragile. Right: both quote wide and neither undercuts. Each takes half the fills at the wide price and earns half the monopoly profit. The difference between the right panel and the left is the rent. The striking fact, from Colliard, Foucault and Lovo and replicated by us, is that independent learning algorithms end up in the right panel without ever communicating. Each learns that undercutting today gets punished tomorrow, and stops. That is tacit collusion, and the rent is what the uninformed taker pays for it.

The puzzle: on the tape, rent and toxicity look identical

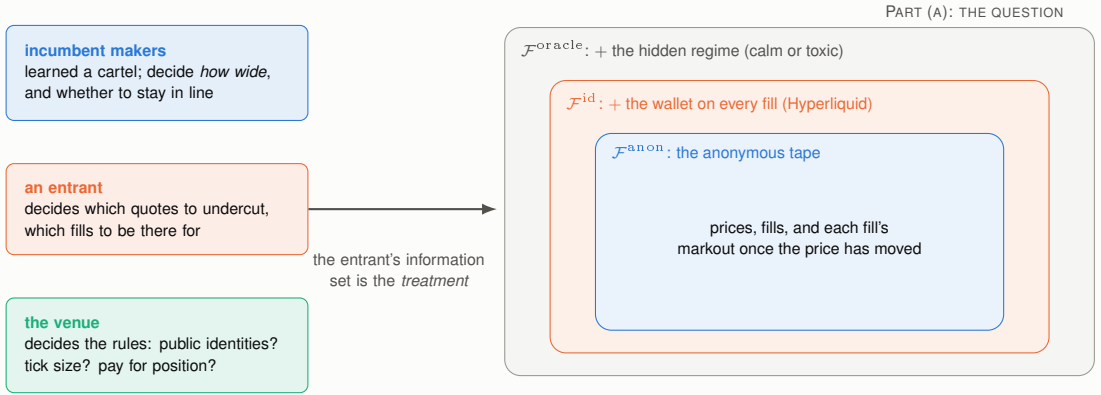


A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Here is the puzzle. This is the maker's expected profit per period as a function of its half-spread, one curve per level of toxicity alpha, the fraction of takers who are informed. Each curve has a break-even point on the left, h-C, and a peak, h-M. More toxic flow pushes the whole curve down and both points to the right. So if you watch a market and see wide quotes, you cannot tell whether the makers are colluding on a calm market, the blue curve, or correctly charging for a toxic one, the red curve. Rent and toxicity produce the same tape. The only ways to tell are to look at outcomes, the markouts, or to look at who is trading. And that second route exists only because wallets are public.

Three players, and what each one gets to know



Who plays. Three players. The incumbents, in blue, are the makers who have already learned a cartel; they decide how wide to quote and, on a one-tick book, whether to stay in the queue. The entrant, in orange, is the adversary: a new maker who decides which quotes to undercut and which fills to be there for. The venue, in green, sets the rules: are identities public, how coarse is the tick, can a maker pay for queue position. The right side is the information structure, nested. The anonymous tape, which every exchange in the world shows: prices, fills, and eventually each fill's markout. Add the wallet on every fill and you have what Hyperliquid shows. Add the hidden regime, whether the market is currently calm or toxic, and you have an oracle that we use only as an upper bound. The entrant's information set is the treatment: we run the same attacker with each of the three and the difference is what identity is worth.

What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

How much of a learned cartel's rent can an adversary take or destroy?

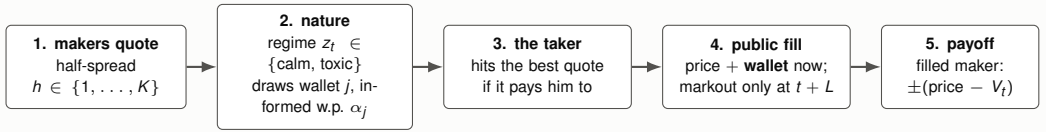
RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: the rent is fully exploitable; identity's value is a lag story; and on the real book the makers who earn are the ones avoiding the toxic wallets.

Three questions. One: what is a public wallet worth to an entrant beyond what the anonymous tape already tells it, and through what mechanism. Two: how much of a learned cartel's rent can an adversary take for itself or destroy, and does a bigger rent mean a bigger target. Three: on Hyperliquid's major books the spread is pinned at one tick by the grid, so the quote cannot carry rent; where does the rent live then, and which venue rules decide who ends up holding the toxic flow. Why this is a game and not an optimisation: the entrant's weapon, undercutting or cherry-picking fills, is exactly what the incumbents' learned reaction function is tuned to punish, and the venue decides what each side is allowed to know. The three headlines today: the rent is fully exploitable; identity's value is a story about lag; and on the real book the makers who earn the spread are the ones who are missing from the toxic wallets' fills.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h;\alpha) = (1-\alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

The model, one period. Step one: every maker commits a half-spread h on the tick grid. Step two: nature draws a regime, calm or toxic, from a persistent Markov chain, then a wallet from a population of fifty; each wallet has its own probability alpha-j of being informed, and in the toxic state the active wallets are the toxic ones. Step three: the taker hits the best quote if it pays him to, which for an informed taker means exactly when the maker will lose. Step four: the fill is public at once, price and wallet, but its markout, the value V-t the maker traded against, is only revealed L arrivals later. That lag L is the key parameter. Step five: the filled maker's payoff is price minus value. One hundred periods make an episode; the regime chain and the wallets persist across episodes. The incumbents are tabular Q-learners whose state is their own last quote and the best rival's; they are identity-blind by construction; we train them twenty thousand episodes and then freeze them, so a cartel is a fixed reaction function. The entrant's information set is the treatment, and a Bayesian filter turns whatever it sees into an estimate of the next taker's alpha. And because the market is Glosten-Milgrom, the expected payoff of a quote against informed share alpha is closed form, which is what makes rent measurable rather than estimated.

Benchmarks are closed-form, so “rent” is measured, not estimated

$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\mathrm{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\mathrm{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

Because the payoff curve is closed form, the benchmarks are exact. h-C is the smallest break-even quote, the Nash outcome of one-shot competition: any wider quote is profitably undercut. h-M is the joint-monopoly quote. The rent index C puts a frozen cartel between zero, Nash, and one, monopoly; the cartels we train sit at 0.7 to 0.9. Under an adversary A, L-B-R is what the cartel loses, and kappa-A is that loss as a share of the rent it had; kappa above one means the attacker destroyed more than the rent. All our results are in rent units, divided by Pi-M minus Pi-C, so cartels at different toxicity levels are comparable. A worked example at alpha 0.3: two incumbents at fourteen ticks split 1.50 per period. An entrant at thirteen takes every fill and earns 1.49, they earn zero. If they chase, the entrant goes to twelve, and the chase ends at h-C equals four, where everyone earns 0.12. A learned cartel is a reaction function that does not chase every time; that restraint is the rent, and it is also the attack surface, because the attacker can learn exactly when it chases and when it does not. The queue mode models the one-tick book: the spread is pinned, so the only decisions are rest, step out, or pay for priority, and the fill queue is first in first out. At break-even a fill is worth plus 1.10 in a calm spell and minus 2.26 in a toxic one, so all the value is in which fills you take.

The prototype, in one command

`python experiments/demo.py`

1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

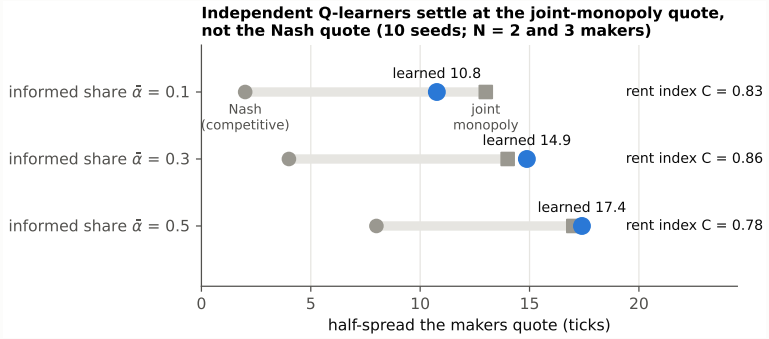
`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)                 0.41           0.33     0.48
  markout (undercut if rent inferred)        0.47           0.41     0.55
  exact best response                        0.94           1.09     1.10

Trace: exact best response vs the cartel (quotes in ticks)
    t  inc0  inc1  entrant   taker          filled   payoff
    0    15    24       12   noise    buy  entrant   +1.21
    1    23    16       13   noise    buy  entrant   +7.86
    2    20    14       13   no trade
    5    21    19       14   informed buy  entrant   -5.77
    7    18    23       14   noise    buy  entrant  +11.79
```


The prototype runs in seconds with one command. It prints the closed-form benchmarks, plays the shipped two-maker cartel alone to show its rent index of 0.86, enters three adversaries under common random numbers, and traces the exact best response period by period. The trace is worth reading: the incumbents answer each entrant quote with their learned reaction, and the entrant sits one tick inside what they are about to quote, holding them in states they never trained on. The package has the simulator, the learners, the entrants, the queue mode, the benchmarks, every experiment, and the Hyperliquid pipeline: recorder, scoring, identity selection. Fifty-nine tests. Every result in the report is logged in REPLICATION-NOTES with the command that produced it.

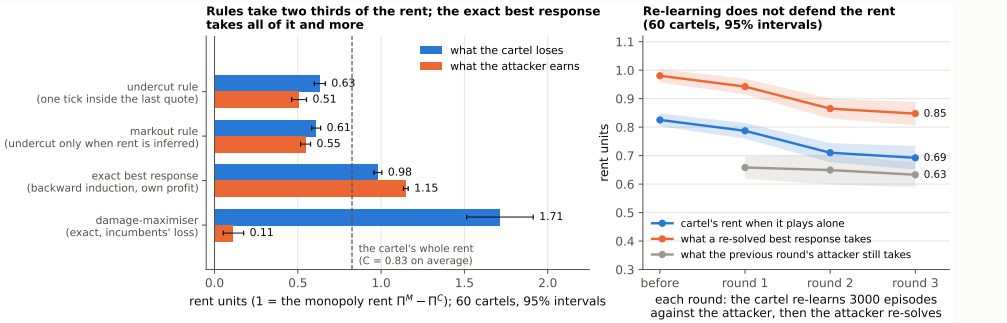
Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

First piece of evidence: the thing we are attacking exists. Independent Q-learners settle at a rent index between 0.71 and 0.87 in every cell of a grid over the number of makers, the learning rate, and the initialisation. That replicates Colliard, Foucault and Lovo. The mechanism is restraint, and we can see it directly: in fifty-five to eighty-three percent of the states the learners visit, undercutting would pay this period, and in ninety-six to one hundred percent of those states the learner does not do it. It has learned that the rival punishes.

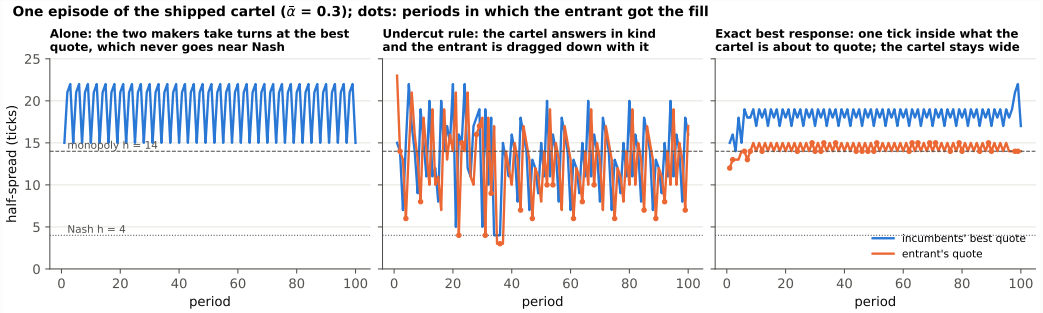
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3. The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

Second: how much of that rent can an adversary take. Two hand-written rules, undercut by one tick and undercut when the markouts say there is rent, take about two thirds, flat across a hundred and eighty cartels. The exact best response is different in kind: because a frozen cartel is a deterministic reaction function, the entrant's problem is a finite-horizon dynamic program we solve by backward induction. It takes more than the whole rent, L-B-R greater than C in sixty of sixty cartels, kappa 1.1 to 1.3, and it earns the monopoly profit for itself. How: it undercuts the reaction, not the last quote, and holds the cartel in states it never trained on, where its learned answer is to quote wider. And a damage-maximiser, the same solver with the objective flipped to minimise the cartel's profit, steers the cartel tight instead, inside break-even, destroying two to three times the rent while taking no fills and spending nothing.

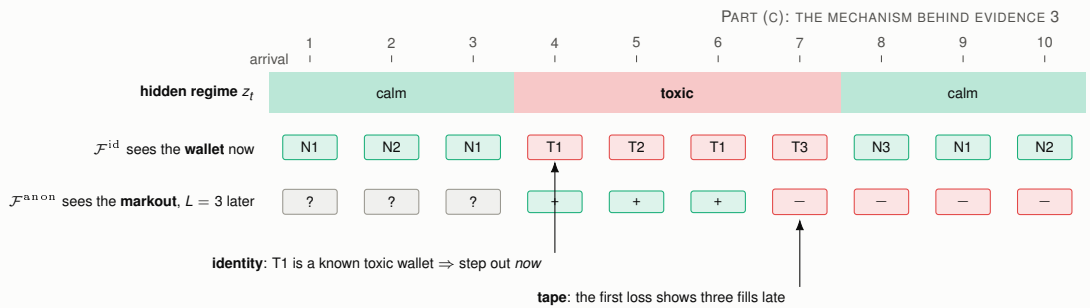
Steering, period by period



Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

What steering looks like. Left: the learned cartel alone is a rotation; the two makers alternate at the best quote, wide, and never go near Nash. Middle: the undercut rule starts a price war; both sides slide toward break-even and the entrant's fills, the dots, come at ever worse prices. Right: the exact best response never provokes the war. It quotes one tick inside the reaction it has predicted; the cartel's learned answer to that is to stay wide; and the entrant takes nearly every fill at about fourteen ticks. The monopoly profit, for itself, with the cartel's own restraint doing the work.

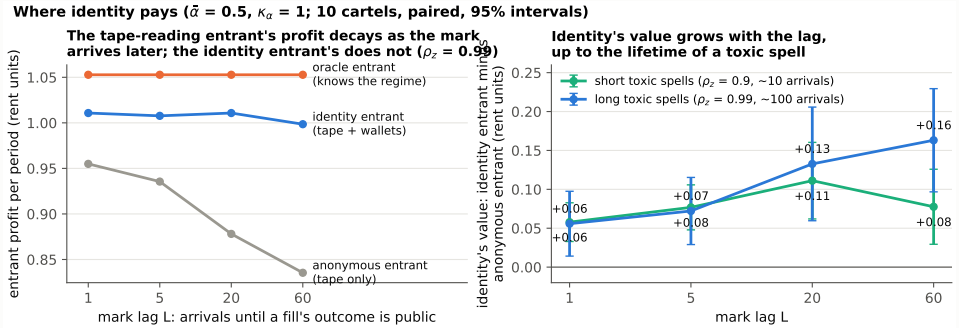
Why identity has value: a spell of toxic flow, seen two ways



Identity is worth what you lose in the L fills before the tape catches up.

Why would a wallet be worth anything beyond the tape? Because informed trading comes in spells, and the toxic wallets are the ones active in a spell. Top row: the hidden regime switches to toxic at arrival four. Middle row: with identities, the fourth fill carries wallet T1, which you have seen lose you money before, so you know the spell has started and you step out now. Bottom row: the anonymous tape only learns a fill's markout L arrivals later; with L equal to three, the first minus sign appears at arrival seven. Three fills late. So identity is worth about what you lose on those L fills: zero if the tape is instant or the spell is harmless, and growing with the lag up to the spell's lifetime, roughly one over one minus rho. On Hyperliquid the analogue of L is real: a fill's outcome is only knowable once the price has moved, seconds to minutes, while the wallet is on the fill at once.

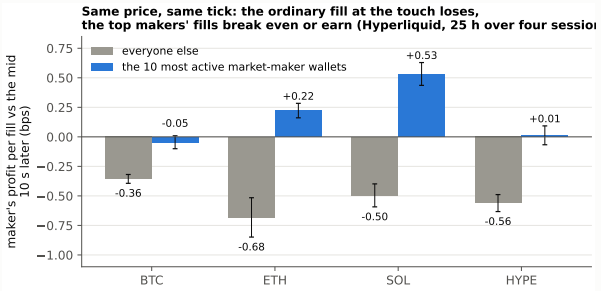
Evidence 3: what identity is worth, and why



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Measured. Our original proposal's mechanism, that the same wallet trades again so you can learn it, is worth nothing on the data: a wallet repeats within the window two to five percent of the time. The mechanism that survives is the lag story from the previous slide. The value of identity is zero wherever even the toxic state leaves an undercut profitable, alpha up to 0.3, because then you do not care who is coming. It becomes positive at alpha 0.5 and grows with the mark lag L, saturating at the spell's lifetime. So RQ1's answer is: identity is a lag story, and its value depends on the venue's reporting delay and the market's toxicity, not on recognising individuals.

Evidence 4, Hyperliquid: the price cannot move, and someone still earns



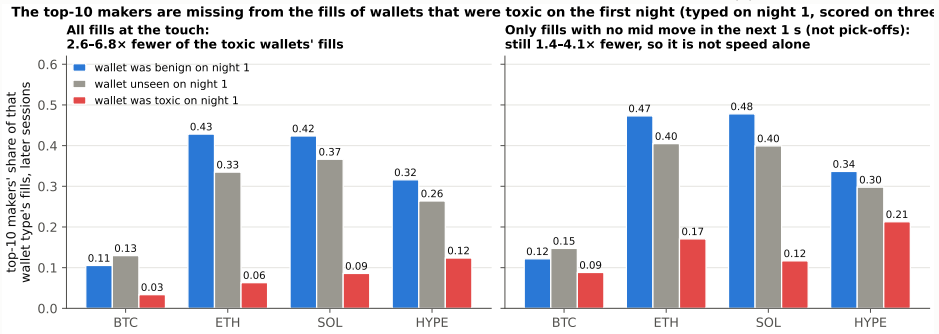
Four sessions, 25 h, 490k scored fills.

- **The spread is one tick**, essentially always
- **The ordinary fill loses**: -0.2 to -0.4 bps in 10 s
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

Rent lives in *which fills you get*, not in the quote.

Now the real venue. We recorded the best bid and ask and every fill with both wallets on Hyperliquid, four sessions over two weeks, twenty-five hours, four hundred and ninety thousand scored fills, plus four hundred and forty-nine hours of fills alone. Three facts. On BTC, ETH and SOL the spread is one tick essentially always; the ninetieth percentile equals the median. Rent cannot live in the price. The ordinary fill loses: a maker filled at the touch collects the tick and then loses 0.2 to 0.4 basis points to the mid within ten seconds. The taker's real cost is in depth: one order in ten walks the book and pays twenty ticks. And the ten most active maker wallets, chosen on activity and scored on outcome, earn 0.3 to 1.0 basis points more per fill at the same price, and their ranking persists from session to session, rank correlation 0.5 to 0.8. So on the real book rent lives in which fills you get, which is exactly the queue-mode model.

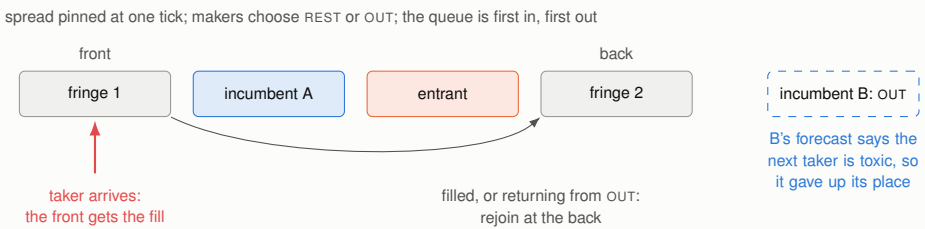
Evidence 5: how they do it is identity, not the tape



They never withdraw after a stressful spell (30 s to 5 min). They are absent from the toxic *wallets'* fills: typed on night 1, scored on three later sessions.

How do the top makers do it? First what it is not: they never withdraw after a stressful spell, at any timescale from thirty seconds to five minutes; the correlation between recent stress and their presence is between minus 0.11 and plus 0.13. They are not reading the tape. What they are doing is identity. We typed every taker wallet on the first night by the markout it inflicted, then on three later sessions over two weeks measured each maker's share of fills against toxic wallets versus benign ones. The top ten makers hold eleven to forty-three percent of fills against benign wallets and three to twelve percent against toxic ones; the gap is the same inside calm and stressed tape, and it survives on quiet fills where the mid does not move, so it is not that they are being picked off. Out of sample: the toxic wallets typed on night one still cost one to two basis points more on the later nights. This is RQ1's mechanism observed on the real book: the makers who earn are the ones missing from the toxic wallets' fills.

The queue game: the one-tick book

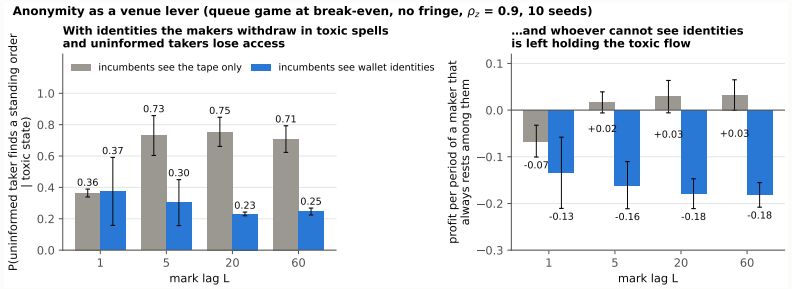


A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays: whoever rests through the toxic spell, the *fringe* (retail orders, the vault).

The model for the one-tick book. The spread is pinned, so each maker's only decision is rest or step out, and those resting stand in a first-in-first-out line; the taker takes the front. A filled maker rejoins at the back; so does one that stepped out and comes back, which is the cost of stepping out. Incumbent B has stepped out because its forecast says the next taker is toxic; it gives up its place in line to avoid that fill. An identity-aware maker does this on the wallet label; a tape-reading one on markouts L late. A fill at break-even is worth plus 1.10 in a calm spell and minus 2.26 in a toxic one, so all the value is in which fills you take, and time priority rations them. Who pays? Whoever rests through the toxic spell: the fringe, which is retail orders and the vault. We also allow rest-and-pay, a priority fee that buys the front of the line, which turns the calm fills into an all-pay auction.

Evidence 6: the levers decide who holds the toxic flow, not whether rent survives



Anonymity: a transfer from makers to uninformed takers in stress; identity the reverse.

Priority fee: the venue collects 2 to 4× the rent; the identity-blind fringe pays.

Makers' own levers: only an identity-aware floor helps, and it leaves the adversary 75 to 80%.

The venue's levers, in the queue game. Anonymity is a transfer: it moves money from makers to uninformed takers in stress episodes, because nobody can step out in front of a toxic wallet; identity moves it the other way. Neither defends the rent. The priority fee is an all-pay auction for position: at the venue's optimum it collects two to four times the makers' rent and leaves every maker at or below zero, and the fringe that cannot see identities is who pays. The makers' own defences, requoting, randomising, a floor on the quote, mostly cost profit and buy nothing; only an identity-aware floor improves both profit and resilience, only where the toxic regime bites, and it still leaves the adversary seventy-five to eighty percent of its take. So RQ3's answer: the rules decide who holds the toxic flow, not whether the rent survives, and the party that ends up holding it is the one that cannot see.

Plan for Workshop 2

Attacks and learning

1. **Public-information attacker**: identify the reaction function from the tape, then solve
2. **Both sides learning**: PSRO; are adversarially trained cartels less exploitable, or just less collusive?
3. **Queue-mode attacks** on no-regret incumbents

Designs, and data

- Venue and maker levers on one frontier: profit vs worst-5% under attack, plus takers' access in stress
- A month of recording, by coin, hour and regime; the order-level archive

Your view: is a frozen reaction function the right "realistic attacker"?

For Workshop 2. On attacks: first, the attacker with public information only. Every incumbent answer to every pair of quotes is on the tape, and the entrant sets the rival's quote with its own, so it can identify the reaction function by probing and then solve it; we will report its gap to the exact best response as a function of the probing budget. Second, both sides learning, PSRO with re-learning cartels and learned entrants: what is the equilibrium rent, and are adversarially trained cartels less exploitable or just less collusive. Third, attacks in queue mode on no-regret incumbents. On designs: venue levers and maker levers on one frontier, profit alone against profit in the worst five percent of episodes under attack, against all three attackers, plus uninformed takers' access in stress so that robust cannot mean withdrew. On data: the recorder has been running continuously since October 25 and will run for a month, so we can cut the identity-selection gap by coin, hour and regime; and the order-level archive tells us which action, cancel, queue position, depth, carries the selection. The question we would most like your view on: is identifying a frozen reaction function the right realistic attacker, or should the cartel be allowed to adapt continuously?

If you remember three things

1 Rent is not defensible

Solve the cartel's reaction function and you take the monopoly profit. Re-learning does not get it back.

2 Identity is a lag story

A wallet reveals a toxic spell before the price does. Worth zero when the tape is instant, more with the lag.

3 The real book agrees

On a one-tick book rent is which fills you get. The makers who earn avoid last night's toxic wallets.

`github.com/pradeepu191/wargame`

Recorder running continuously since October 25.

Three things. One: the rent is not defensible. An entrant that solves the cartel's reaction function takes the monopoly profit for itself, and in a double oracle where the cartel re-learns against it, the re-solved attacker takes it again. Two: identity's value is a lag story. A wallet reveals a toxic spell before the price does; that is worth zero when the tape is instant or the spell is harmless, and grows with the reporting lag. Three: the real book agrees. On a one-tick book the rent lives in which fills you get, and the makers who earn are the ones missing from last night's toxic wallets' fills, out of sample, calm or stressed. Code, results and a findings log are on GitHub. Thank you.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.